Epistemic Case Study — Findings, with Data

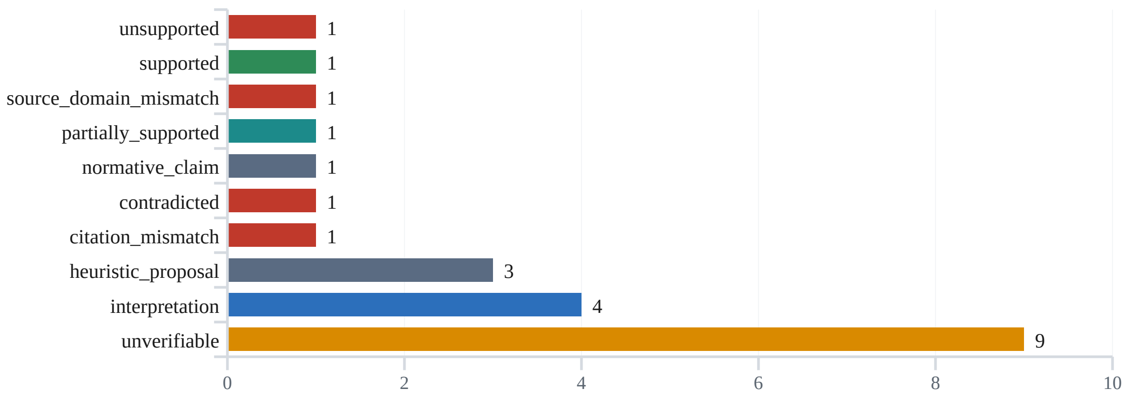
The “epistemic validation” of a Muse Spark 1.1 text by MarCognity-AI, analysed with DESi

Source: Hugging Face post *“Epistemic Stress Test — Muse Spark 1.1 validated by MarCognity-AI”* (user elly99, 2026-07-10; Zenodo 10.5281/zenodo.20509721) and github.com/elly99-AI/MarCognity-AI. Regenerated offline & deterministically with `python -m desi.case_studies.marcognity_muse_spark`. Every number below comes from `summary.json / claims.jsonl / evidence.jsonl`.

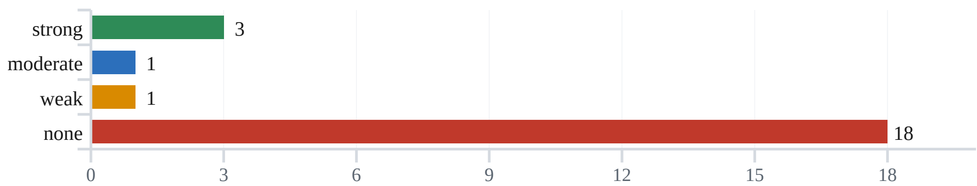
23 atomic claims typed & anchored	4/23 claims with domain-admissible evidence	3 structural conflicts (1 logical contradiction)	yes self-sealing conclusion; falsifier: no

Distribution of verdicts (23 claims)

No binary VERIFIED/FAILURE stamps: heuristics, interpretations and normative statements get their own categories; “unverifiable” is a first-class verdict, not a fallback.



Evidence strength — where concrete support is missing



18 of 23 claims have no concrete evidence passage at all — the validator referred throughout to “the PubMed document” with no title or locus. (Audit R4: these counts describe the experiment's supplied evidence over 23 curated claims, not a measured groundedness of the Muse text.)

The three structural conflicts

Found by DESi's own detector `desi.self_audit.contradictions.find_contradictions` as key/value inconsistencies — not asserted by prose. Not every one is a logical contradiction; the label names its strength.

C1 — Logical contradiction: Prompt ↔ Method

The Method (muse:L206): “No epistemic instructions (verification, sources, stages)”. The printed prompt demands ≥5 references with direct citations (L56–58), a citation-consistency check (L64) and six named Phases (L29–47). A genuine contradiction: both cannot be true at once. (Audit caveat R1: the prompt header muse:L12 reads 'prompt used on chatgpt (taken from marcognity)', so its provenance is slightly ambiguous; the contradiction holds on the Method's own wording.)

C2 — Pipeline inconsistency: VERIFIED without citable evidence

“VERIFIED” and “No citations found” can in principle both hold — this is not a strict logical contradiction. The real fault is an unresolved pipeline inconsistency: the Skeptical Agent asserts source support (L170–198) but names no source and no passage (only “the PubMed document”); the citation checker meanwhile finds no verifiable references (L201–202) — although the text carries eight (L154–161). Both results are merged without a consistency check (agent_metacognition L48–66).

C3 — Unsubstantiated independence: “independent” ↔ no documented independence

That the validator is technically a single `llm.invoke` (skeptical_agent L62) does not by itself refute independence — an external LLM call could be formally independent. The conflict is narrower: “independent external validation” (L208) is claimed with no documented independence on any axis — organizational, model-side or evidence-side. The validator receives exactly the documents that fed generation (evaluator_prompt L24–28), is non-adversarial and not reproducibly specified. Asserted, not established — a methodological misclassification.

Self-sealing & falsifiability

Would support	Validator flags the unverifiable → the gap is made visible; OR the validator fails → the Boundary “within the validator itself” (L237). Both outcomes confirm.
Would weaken	A pre-registered run in which the validator, given domain-correct sources, confirms claims with quoted passages — not provided.
Would refute	A control in which the residual failures vanish under proper source-gating (a pipeline defect, not an “intrinsic architecture”) — not provided.
Falsification condition stated?	**No.** Since success and failure both confirm and no outcome is designated disconfirming, the hypothesis is unfalsifiable *as run*.

All 23 claims — the data

A **curated selection** (incl. previously missed claim classes), **not** a measured complete claim coverage of the Muse text. Line-by-line auditable in `claims.jsonl` / `evidence.jsonl`. Column “Evidence” = provenance kind of the source actually available.

ID	Type	Domain	Verdict	Evidence
MET-01	methodological	experiment methodology	contradicted	primary
MET-02	methodological	experiment methodology	partially_supported	primary
MET-03	methodological	experiment methodology	unverifiable	none
VAL-01	claim about validator	legal philosophy	source_domain_mismatch	semantic-only
VAL-02	claim about validator	experiment methodology	citation_mismatch	primary
VAL-03	claim about validator	experiment methodology	supported	primary
EB-01	claim about epistemic boundary	ml epistemology	interpretation	none
EB-02	claim about epistemic boundary	ml epistemology	unsupported	secondary
DEF-01	definition	legal philosophy	interpretation	none
DEF-02	definition	legal philosophy	interpretation	none
ATTR-01	author attribution	legal philosophy	unverifiable	none
ATTR-02	author attribution	legal philosophy	unverifiable	none
QUOTE-01	direct quote	legal philosophy	unverifiable	none
QUOTE-02	direct quote	legal philosophy	unverifiable	none
FACT-01	historical fact	legal philosophy	unverifiable	none
HEUR-01	heuristic model	legal philosophy	heuristic_proposal	none
HEUR-02	heuristic model	legal philosophy	heuristic_proposal	none
HEUR-03	heuristic model	legal philosophy	heuristic_proposal	none
THEO-01	theoretical interpretation	economics	interpretation	none
EMP-01	empirical	positive law	unverifiable	none
FACT-02	juristic fact	positive law	unverifiable	none
FACT-03	historical fact	technology history	unverifiable	none
NORM-01	normative	legal philosophy	normative_claim	none

MarCognity vs. DESi — what differs

Not a sales pitch: DESi does not prevent errors, it makes visible the point at which an error, an omission or an inadmissible inference arises.

Dimension	MarCognity (Skeptical Agent)	DESi
Claim coverage	5 general claims	23 curated, typed claims (incl. missed classes); not measured completeness
Source fit	none (PubMed ↔ law)	source_domain_gate → mismatch, not VERIFIED
Concrete provenance	“the PubMed document”	exact anchor (doc:line) or none
Contradiction detection	misses C1, produces C2	C1/C2/C3 via find_contradictions
Interpretation/heuristic	binary VERIFIED/FAILURE	heuristic_proposal / interpretation / normative
Uncertainty	global prose	per claim: verdict + uncertainty + strength
Falsifiability	no condition	names support/weaken/refute + missing falsifier
Auditability	one concatenated text	jsonl per line + optional hash-ledger
Evaluator self-check	none; failure = confirmation	the report is itself under test (VAL-01..03)

Core finding

The demonstration is not that Muse Spark makes mistakes. It is: the claimed validation **confirms general legal statements with unsuitable, non-transparent sources** (PubMed for legal philosophy, without title/passage), **overlooks a direct contradiction in the setup** (C1) and then **reads its own failure as confirmation of the theory** (self-sealing, with no falsification condition).

Limits of DESi (honestly)

DESi is not infallible: the claim fixture is a curated selection (no auto-extractor), **not measured complete coverage** of the Muse text; the legal philosophy is *not* adjudicated on the merits here (many claims deliberately end on “unverifiable”); source_domain_gate and the self-sealing analysis are small, general extensions. MarCognity's own README/Boundary document are more careful than the forum conclusion — the over-reach is in the conclusion, not in the hedged hypothesis. In fairness (R5), MarCognity's approach has real strengths — claim decomposition, multi-source retrieval, an explicit skeptical pass; the fault is in gating and provenance, not the idea.

Part II — Doktores rule-guided self-audit

A rule-guided **self-audit**: four source-anchored reviewers (provenance, methodology, logic, fairness) attacked the DESi analysis — deterministic, offline, no LLM. Most findings survived, some were qualified, C2/C3 were confirmed as NON-contradictions, dissent is preserved.

Scope of independence: this is **logically and provenance adversarial** but **not organizationally or model-side independent** — all four reviewers are fixed rules from the same repository. A clean deterministic counter-check, NOT an independent replication.

This attestation does NOT certify the truth of the legal statements. It rates only the method, provenance, consistency and reach of the DESi case study.

23	0	2	4
claims reviewed	verdicts overturned by this audit	conflicts reclassified	wording revisions

Transparent balance across the whole review process

- 23 claims reviewed
- 22 confirmed unchanged or with a qualification
- 1 claim verdict revised (MET-02: contradicted → partially_supported; in the operator review before this audit, confirmed here)
- 2 conflict classifications revised (C2 → pipeline inconsistency, C3 → unsubstantiated independence)
- 4 wording revisions by this audit (R1/R2/R4/R5)
- 0 claims fully rejected

The three conflicts under adversarial review

C1	upheld — logical contradiction	prompt ↔ method; both cannot be true (with a small prompt-provenance caveat, R1)	✓
C2	reclassified → pipeline inconsistency	VERIFIED and 'no citations' can co-occur; not a logical contradiction	→
C3	reclassified → unsubstantiated independence	one LLM call does not refute independence; it is simply not documented on any axis	→

Attestation — twelve dimensions, separately rated (no blanket seal)

Dimension	Rating
Reproducibility	passed
Provenance	passed
Claim derivation	with qualifications
Claim coverage	with qualifications
Source fit	with qualifications
Verdict logic	passed
Conflict classification	with qualifications
Falsifiability	passed
Fairness to MarCognity	with qualifications
DESi over-reach	with qualifications
Auditability	passed
Open objections	n/a

Overall attestation: **passed_with_qualifications**. Most findings survived because the artifact was already tight; the audit's teeth are in the reclassifications (C2/C3) and the qualifications (R1/R2/R4/R5), not in a rubber stamp.

Open questions (missing original data)

• The live Hugging Face thread and full MarCognity repo were not re-fetched — only their reproduced content is anchored.

• The actually retrieved 'PubMed document' text is unknown; the domain mismatch stays probable, not certain.

• Muse Spark's version / parameters are unknown; the underlying experiment's reproducibility is not assessable.

• The 23 claims are not measured coverage; the true claim coverage of the Muse text is unknown.